

Introduction

Multicollinearity happens when two or more independent variables are highly correlated with each other. Instead of each feature contributing unique information, they start overlapping. This creates problems in interpreting the models.

For example, for property sales dataset having following columns,

- Floor size in sqft.
- Floor size in sqm.
- No of room
- Location

When floor size in sqft increase the sqm also increases. Also, typically when floor size increases the number of rooms also increase. This is type of **structural multicollinearity**. This will confuse the model that which input variable more influences the prices.

In other hand, two different locations may have the same price. This is an example of **data-multicollinearity**. This also disturb the model to predict.

How to overcome

Variance Inflation Factor (VIF) is one of the common methods used to solve the multiplication problem.

$$VIF = \frac{1}{1 - R^2}$$

If the VIF value,

- 1 = No correlated
- (1 – 5) = Moderately correlated and acceptable
- > 5 = Highly correlated (need action to handle the correlated columns before train the model)